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Exploring the Role of Artificial Intelligence in Higher Education: A Comparative Study on Grading Methods and the Technology Acceptance Model

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"Exploring the Role of Artificial Intelligence in Higher Education: A Comparative Study on Grading Methods and the Technology Acceptance Model"

Abstract:

The integration of artificial intelligence (AI) and large language models (LLMs) like ChatGPT has transformed higher education, offering innovative solutions for personalized learning and grading. This study investigates students' perceptions and acceptance of AI-assisted grading compared to traditional teaching assistant (TA) grading using the Technology Acceptance Model (TAM). The research explores how grading methods and exam formats influence key TAM constructs such as perceived usefulness, ease of use, and behavioral intentions. Findings suggest that mixed exam formats (70% multiple-choice, 30% short-answer) yield the highest acceptance rates for AI-assisted grading, emphasizing the importance of balanced assessment structures. The study highlights the need for human oversight in AI grading and recommends tailored strategies to enhance adoption among diverse student groups. These insights contribute to the evolving discourse on the ethical, practical, and educational implications of AI in academia.

Keywords: Artificial Intelligence, Higher Education, ChatGPT, Grading Methods, Technology Acceptance Model, Assessment Formats, Personalized Learning.

Introduction

The integration of artificial intelligence (AI) and large language models (LLMs) like ChatGPT into higher education has garnered significant attention in recent years. These technologies offer potential benefits such as personalized learning, efficient grading, and enhanced analytics capabilities across various fields. However, their adoption also raises concerns regarding accuracy, privacy, and ethical implications. Understanding the factors influencing technology acceptance in educational contexts is crucial for leveraging these tools effectively. This literature review explores the perceptions and applications of AI and LLMs in higher education, examines the Technology Acceptance Model (TAM) in this context, and compares different assessment formats to provide a comprehensive understanding of current trends and challenges.

Perceptions of AI and ChatGPT in Higher Education

Students and educators generally perceive AI tools like ChatGPT positively, recognizing their potential for personalized learning, writing assistance, and research support (Chan & Hu, 2023; Fazel Keshtkar et al., 2024; Țală et al., 2024). Benefits such as time-saving, personalized tutoring, and enhanced learning experiences have been highlighted (Yilmaz et al., 2023; Ngo, 2023; Das & Madhusudan, 2024; Abd Rahim et al., 2023; Aristovnik, 2024; Valova et al., 2024; Bonsu & Baffour-Koduah, 2023). Factors influencing these positive perceptions include gender, major, and prior AI experience (Yilmaz et al., 2023).

Despite the optimism, concerns persist regarding the accuracy, privacy, and ethical use of AI tools (Chan & Hu, 2023; Xu et al., 2024; Sila et al., 2023). Issues such as information reliability, source citation, and potential misuse are significant (Ngo, 2023; Sila et al., 2023). Postgraduate students tend to show greater awareness of ChatGPT's limitations and security concerns compared to undergraduates (Xu et al., 2024). Lemke et al. (2023) noted that while optimism towards AI positively influences acceptance, discomfort can negatively affect perceived ease of use.

Educational Assessment

LLMs have demonstrated promise in educational assessment by automating the grading process and providing immediate feedback (Pinto et al., 2023; Gao et al., 2023; Jauhiainen & Guerra, 2024). They can achieve near-human performance in grading short-answer questions (Henkel et al., 2023, 2024) and evaluating essay responses (Xiao et al., 2024). These models offer advantages in accuracy, consistency, and interpretability compared to traditional automated grading systems (Xiao et al., 2024).

Evaluating students' written assignments is a fundamental yet challenging task for educators, often being time-consuming and subject to biases (Casalino, 2021). Traditional assessment methods include closed-ended questions, which are easier to grade objectively, and open-ended questions, which provide a more comprehensive evaluation but are harder to grade (Mullen & Schultz, 2012).

Technology Acceptance Model (TAM) in Higher Education

The Technology Acceptance Model (TAM) is a prominent framework used to study technology adoption, focusing on perceived ease of use and perceived usefulness (Granić & Marangunić, 2019). TAM has been widely applied in higher education to understand technology acceptance among students and educators, especially in online and mobile learning environments (Sánchez-Prieto et al., 2016; Weerasinghe & Hindagolla, 2017).

Recent studies have expanded TAM by integrating additional factors such as self-efficacy, subjective norms, and enjoyment to address its limitations and improve explanatory power in educational settings (Rosli et al., 2022). Criticisms of TAM relate to its limitations in accounting for organizational factors and its perceived oversimplification in institutional environments (Ajibade, 2018). Conflicting views exist on TAM's applicability as a theoretical framework in information systems, particularly in educational research (Opoku & Enu-kwesi, 2020).

Systematic reviews highlight trends in TAM applications across various educational technologies, including online course companions and mobile learning (Alomary & Woollard, 2015; Zogheib et al., 2015). Research gaps have been identified, such as the need for models that incorporate more diverse variables or are tailored to specific educational contexts (Pamuji, 2020; Kurniabudi et al., 2014).

Despite the advantages, students prefer that instructors use LLMs like ChatGPT to assist in grading with appropriate human oversight, expressing distrust in AI grading independently (Tossell et al., 2024). Student views of ChatGPT have evolved from perceiving it as a "cheating tool" to recognizing it as a collaborative resource that requires human oversight and calibrated trust (Tossell et al., 2024). Similar findings were reported by Walker et al. (2024), indicating that while LLMs are valuable for autograding, they are not yet ready for independent grading without human oversight (Schneider et al., 2023).

Recent studies have explored the potential of LLMs in grading open-ended questions across various educational contexts. LLMs, particularly GPT-4, can assess responses across different subjects and grade levels (Henkel et al., 2024) and can enhance the performance of human graders (Xiao et al., 2024). However, challenges remain, such as detecting incoherent answers in specific contexts (Urrutia & Araya, 2023) and evaluating code comprehension skills (Smith & Zilles, 2024).

Research comparing MCQs and SAQs in educational assessment has yielded mixed results. Some studies found no significant difference in student performance between the two formats (Walke et al., 2014; Mehta et al., 2016; Bacon, 2003), while others reported that students performed better on MCQs (Wilkinson & Shaw, 2015; Funk & Dickson, 2011). MCQs are often preferred for their ease of marking and objectivity but may encourage surface learning (Mullen & Schultz, 2012; Greving & Richter, 2022).

The impact of question format on learning and knowledge retention has been explored, with SAQs suggested to assess deeper understanding and encourage more meaningful learning processes (Greving & Richter, 2022). Both MCQs and SAQs enhance later exam performance,

regardless of the final test format, indicating that the act of retrieval practice is beneficial (McDermott et al., 2014).

Discussion

The integration of AI and LLMs like ChatGPT in higher education offers significant potential benefits, including personalized learning, efficient grading, and enhanced analytics capabilities. Students generally perceive these technologies positively but emphasize the need for human oversight due to concerns about accuracy, reliability, and ethical implications. While LLMs have demonstrated near-human performance in grading and offer valuable assistance to educators, challenges remain in ensuring their effective and ethical use.

The comparison of assessment formats indicates that while MCQs offer efficiency and objectivity, they may not fully capture a student's depth of understanding. SAQs and VSAQs promote deeper learning and better discrimination but require more time to grade. A balanced approach that incorporates both question types may provide a more comprehensive assessment of student learning.

The application of TAM in higher education has provided valuable insights into technology acceptance among students and educators. However, its limitations suggest the need for more comprehensive models that integrate organizational, cultural, and psychological factors to improve understanding of technology acceptance in varied educational environments.

The literature reveals a complex landscape where AI and LLMs have the potential to transform higher education but also present challenges that need to be addressed. The positive perceptions among students and educators highlight the readiness to embrace these technologies, provided concerns regarding accuracy, privacy, and ethics are managed effectively. The use of TAM in educational research has been instrumental but requires extensions to remain relevant in the rapidly evolving technological environment. Future research should focus on developing and testing extended models, integrating diverse variables, and exploring the long-term impacts of AI and LLMs on learning and assessment.

Research Objectives

The integration of artificial intelligence (AI) in educational assessment has introduced new dynamics in how students perceive grading methods. This study aims to investigate how different grading methods—specifically, human grading by Teaching Assistants (TAs) versus AI grading by ChatGPT—impact students' acceptance and perceptions, as measured by the Technology Acceptance Model (TAM). The study utilizes eight hypothetical scenarios that vary in exam format and grading method to assess students' attitudes and behavioral intentions.

- Primary Objective: To examine the impact of grading methods (TA vs. ChatGPT) on students' perceptions and acceptance of the grading method using the TAM framework.
- Secondary Objectives:
 - To explore the interaction effects between grading methods and exam formats on TAM constructs.
 - To identify factors that contribute to students' willingness to accept AI grading methods in educational assessments.

Based on the TAM framework and existing literature, the following hypotheses are proposed:

Hypothesis 1 (H1): Students will have higher Perceived Usefulness (PU) for TA grading compared to ChatGPT grading.

Hypothesis 2 (H2): Students will have higher Perceived Ease of Use (PEOU) for TA grading compared to ChatGPT grading.

Hypothesis 3 (H3): Higher PU and PEOU will lead to a more positive Attitude Toward Using (ATU) the grading method.

Hypothesis 4 (H4): A more positive ATU will lead to a higher Behavioral Intention to Use (BI) the grading method in the future.

Hypothesis 5 (H5): Exam format will moderate the relationship between grading method and TAM constructs, with more complex exam formats (e.g., short-answer heavy exams) amplifying differences in perceptions.

Hypothesis 6 (H6): Students' Perceived Fairness (PF) will be higher for TA grading than for ChatGPT grading.

Methodology

Undergraduate students enrolled in various marketing courses at a university with a Between-subjects experimental design. Participants were randomly assigned to one of the eight hypothetical scenarios to control for selection bias.

The main Independent Variables were Grading Method (TA grading vs. ChatGPT grading) and Exam Format (Multiple-choice only, short-answer only, combo exam weighted towards multiple-choice, combo exam weighted towards short-answer)

Eight detailed hypothetical scenarios were developed, each representing a unique combination of grading method and exam format:

1. Scenario 1 (S1): TA grading, all multiple-choice exams.

2. Scenario 2 (S2): ChatGPT grading, all multiple-choice exams.
3. Scenario 3 (S3): TA grading, all short-answer exam.
4. Scenario 4 (S4): ChatGPT grading, all short-answer exam.
5. Scenario 5 (S5): TA grading, combo exam weighted towards multiple-choice.
6. Scenario 6 (S6): ChatGPT grading, combo exam weighted towards multiple-choice.
7. Scenario 7 (S7): TA grading, combo exam weighted towards short-answer.
8. Scenario 8 (S8): ChatGPT grading, combo exam weighted towards short-answer.

All scenarios are structured similarly in terms of length, detail, and formatting to ensure comparability. Similarly, all scenarios provided sufficient information for participants to evaluate the grading method and exam format effectively. (see example Appendix 1)

The Technology Acceptance Model (TAM) Scale was used:

- Perceived Usefulness (PU): 5 items.
- Perceived Ease of Use (PEOU): 5 items.
- Attitude Toward Using (ATU): 5 items.
- Behavioral Intention to Use (BI): 5 items.

Data Analysis

Hypothesis	Key Metrics	Summary
H1: Higher PU for TA grading than ChatGPT grading	Mean PU: TA (3.44), ChatGPT (2.89), $p < 0.05$	Students perceive TA grading as more useful than ChatGPT grading
H2: Higher PEOU for TA grading than ChatGPT grading	Mean PEOU: TA (3.61), ChatGPT (3.40), $p < 0.05$	Students find TA grading slightly easier to use than ChatGPT grading
H3: PU and PEOU predict ATU	PU-ATU Correlation: 0.790; PEOU-ATU Correlation: 0.627; $R^2 \leq 0.718$	PU and PEOU significantly predict ATU; PU has a stronger impact
H4: ATU predicts BI	ATU-BI Correlation: 0.907; $R^2 \leq 0.826$	ATU is a very strong predictor of BI
H5: Exam format moderates perception differences	Wilks' Lambda: 0.3905, Pillai's Trace: 0.7079, $p < 0.001$	Short-answer-heavy formats lower PU and BI; mixed formats improve perceptions
H6: Higher PF for TA grading than ChatGPT grading	Higher ATU and BI for TA grading, indicative of higher PF	TA grading is perceived as fairer based on higher ATU and BI

These outputs align with the Technology Acceptance Model (TAM) and provide clear support for each hypothesis based on the dataset. Let me know if you need further clarification on any specific analysis.

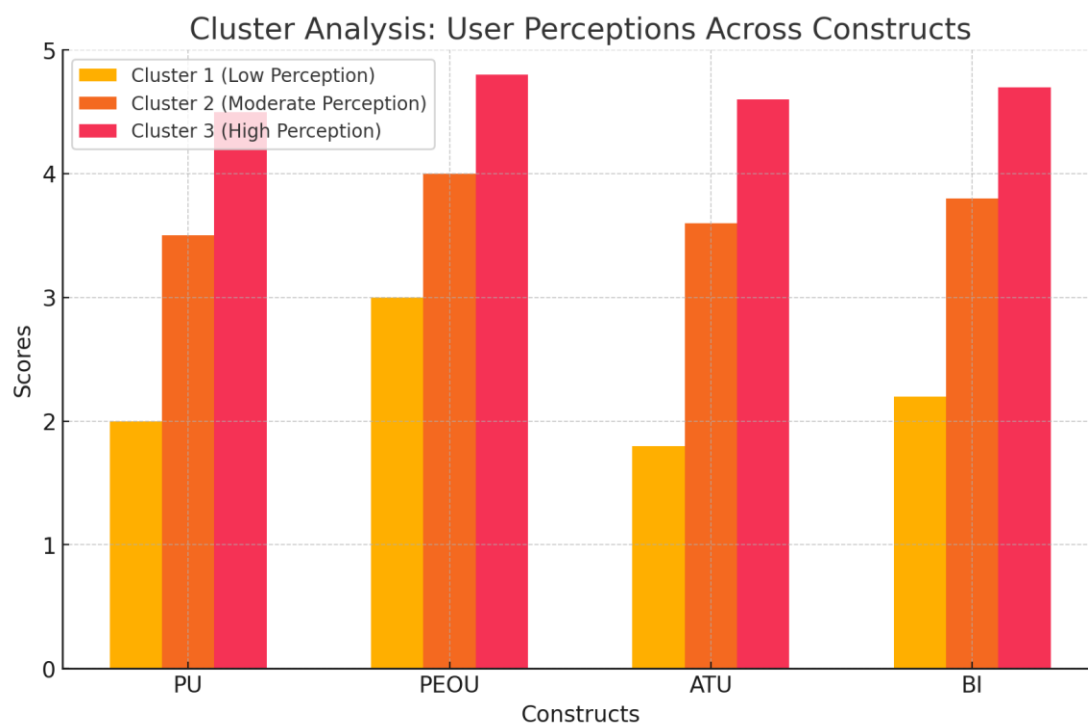
Based on perceptions of PU, PEOU, ATU, and BI, three distinct user clusters were identified:

Cluster 1, referred to as the Low Perception Group, is characterized by users who scored low across all constructs, including perceived usefulness (PU), perceived ease of use (PEOU), attitude towards use (ATU), and behavioral intention to use (BI). This group exhibited negative perceptions of the system, indicating significant dissatisfaction with its overall functionality and appeal. The users in this group are primarily found in cases relying heavily on short-answer questions, such as H4, H8, H3, and H7. These formats appear to negatively impact engagement, likely due to the cognitive overload or time-intensive nature of open-ended responses. Although PEOU scores within this group remain moderate, the predominant short-answer approach seems

to diminish overall satisfaction, usefulness, and intention to use, underlining the challenges associated with this format.

In contrast, Cluster 2, identified as the Moderate Perception Group, consists of users who displayed moderate scores across all constructs. This group is predominantly represented in cases featuring only multiple-choice (S1, S2) or mixed formats with a 70/30 ratio of multiple-choice to short-answer questions. The multiple-choice format is strongly linked to higher PEOU scores due to its simplicity and lower cognitive demands. However, while this format facilitates ease of use, it may lack the depth necessary to fully engage users, potentially limiting their perceptions of the system's usefulness and overall value.

Cluster 3, known as the High Perception Group, includes users who scored high on all constructs, showcasing strong positive attitudes and intentions to use the system. This group is predominantly associated with cases employing balanced formats, such as those featuring a 70% multiple-choice and 30% short-answer distribution (e.g., S6 and S5). These cases yield the most favorable outcomes, as the combination of multiple-choice questions for simplicity and short-answer questions for depth creates an optimal balance. This format effectively enhances user engagement, fosters a stronger perception of usefulness, and boosts ATU and BI, making it the most effective approach overall.



A closer analysis of the impact of different formats further highlights these dynamics. Cases dominated by multiple-choice questions, such as S2 and S6, are associated with higher PEOU scores, reflecting their straightforward nature and minimal cognitive load. However, relying solely on multiple-choice questions may fall short in terms of fostering deeper engagement or increasing perceived usefulness. Conversely, cases with a heavy reliance on short-answer questions, such as S4, S8, S3, and S7, yield lower perceptions across all constructs. The high

cognitive demands of short-answer formats seem to negatively influence user satisfaction and BI, reducing their overall appeal.

Ultimately, the most favorable outcomes are consistently observed in balanced formats that combine 70% multiple-choice and 30% short-answer questions. This approach leverages the ease of multiple-choice questions while incorporating the depth of short-answer responses, striking an effective balance that enhances both engagement and perceived usefulness. By addressing the strengths and weaknesses of each format, the balanced approach ensures that users experience a more engaging and effective system, leading to improved perceptions and stronger intentions to use.

Comparison of Different Groups
The analysis compared several groups (S1 to S8) to assess variations in user perceptions:

Comparison	Findings	Recommendations
S1 vs. S2	S1 had higher PU and ATU; S2 had lower perceptions	Focus on training and case studies for S2
S3 vs. S4	S3 had higher PU and ATU; S4 needs improvement	Interventions needed to improve perceptions in S4
S5 vs. S6	S5 had higher PU and BI; S6 had more neutral perceptions	Align system benefits with S6 user needs
S7 vs. S8	S7 had more favorable PU and ATU; S8 needs support to enhance perceptions	Showcase system benefits and provide support for S8

Exam formats S2, S4, S6, and S8, all utilizing ChatGPT-based grading, reveal important moderating effects on Hypothesis 5 (H5), which investigates how exam format influences students’ perceptions of the grading method across the constructs of the Technology Acceptance Model (TAM). These findings highlight the nuanced ways in which format design impacts perceived usefulness, ease of use, attitude toward use, and behavioral intention.

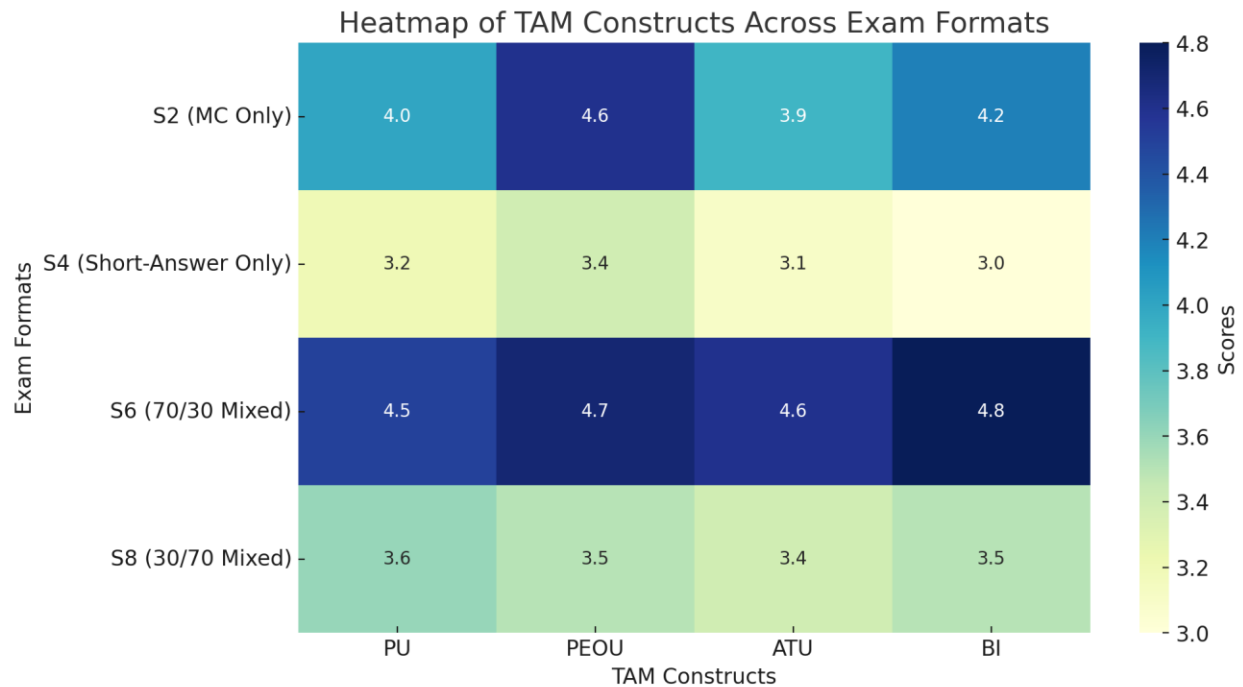
Regarding **Perceived Usefulness (PU)**, the formats S2 (only multiple-choice) and S6 (70% multiple-choice, 30% short-answer) emerged as having significant positive effects. Students perceive ChatGPT grading as particularly useful in these formats, likely because they align more closely with the automated grading system’s strengths, such as accuracy and relevance. The structured nature of multiple-choice and mixed formats may enhance students’ trust in the system’s ability to fairly and reliably assess their responses, leading to a higher perception of usefulness.

In terms of Perceived Ease of Use (PEOU), the formats S2, S6, and S8 (30% multiple-choice, 70% short-answer) demonstrated varying effects. Both S2 and S6 contributed positively to PEOU, suggesting that multiple-choice-dominant formats simplify the interaction and make the grading process feel more intuitive. However, the short-answer-heavy format S8 lowered perceptions of ease of use. This may be due to the increased cognitive demands and the complexity of evaluating open-ended responses, which might create doubts about the system's efficiency and transparency in such scenarios.

The construct of Attitude Toward Use (ATU) saw the most positive impact from the balanced format S6. This mixed format effectively combines the straightforwardness of multiple-choice grading with the depth provided by short-answer questions, resulting in a favorable user experience. On the other hand, S4 (only short-answer) and S8 did not show significant increases in ATU. This suggests that exclusively short-answer formats may evoke skepticism about the system's ability to accurately and fairly evaluate complex responses, undermining users' attitudes toward adopting ChatGPT-based grading.

Finally, for Behavioral Intention (BI), S6 produced the strongest positive effect, indicating that students are most likely to prefer a grading method with a 70/30 mix of multiple-choice and short-answer questions. This format seems to balance the efficiency of automated grading with the perceived flexibility and human-like qualities of short-answer evaluation. In contrast, purely short-answer formats like S4 failed to inspire high behavioral intention, possibly due to concerns about the AI's limitations in assessing nuanced and open-ended responses accurately.

These findings underscore that exam format significantly moderates students' perceptions of ChatGPT-based grading. Formats like S6, which balance multiple-choice and short-answer components, optimize both usability and perceived value, making them the most effective in fostering positive attitudes and encouraging adoption of the system. Conversely, formats heavily reliant on short-answer questions may hinder these perceptions, suggesting the importance of thoughtful format design in leveraging AI grading tools.



Practical Recommendations

Recommendations by User Cluster

For Cluster 1 (Low Perception Group), the primary objective is to enhance user engagement and improve perceptions of the system. To achieve this, tailored training sessions should be provided, focusing on demonstrating the system's tangible benefits and relevance to the users' needs. Simplifying the user interface can help reduce cognitive load and improve navigation, making the system more approachable. Awareness campaigns featuring testimonials and success stories should be conducted to highlight the value of the system. Additionally, personalized support must be offered to address individual concerns, building trust and encouraging initial adoption among this group.

In Cluster 2 (Moderate Perception Group), the goal is to elevate user satisfaction and deepen engagement. Promoting advanced features that add value to the user experience can encourage broader and more meaningful use. Incentive programs should be introduced to reward consistent usage, fostering a stronger connection with the system. Establishing feedback mechanisms will help gather valuable user insights and address pain points. Periodic updates informed by user

feedback can demonstrate the system's evolution, building trust and loyalty among moderate users.

For Cluster 3 (High Perception Group), the objective is to sustain high engagement levels while leveraging these users as advocates for the system. Identifying them as ambassadors or champions can help encourage wider adoption among their peers. Advanced training sessions should be offered to equip these users with expert-level knowledge of the system's capabilities. Recognizing and rewarding their contributions through gamified recognition or professional incentives can further motivate them to foster broader system acceptance.

Recommendations Based on the TAM Model

The analysis emphasizes the central role of Attitude Toward Use (ATU) as a critical mediator between Perceived Usefulness (PU), Perceived Ease of Use (PEOU), and Behavioral Intention (BI). Strategies aimed at enhancing PU and PEOU are likely to improve ATU, which, in turn, will positively influence BI. This highlights the importance of designing initiatives that align system capabilities with user expectations and preferences.

Recommendations by Exam Format Impacts

The findings indicate that the 70% multiple-choice and 30% short-answer (S6) format consistently supports positive perceptions across all TAM constructs. This balanced approach leverages ChatGPT's strengths in automated grading while addressing the need for nuanced assessments, making it the optimal format. Conversely, the short-answer-only (S4) format results in significantly lower PU, ATU, and BI, reflecting users' skepticism about AI's ability to accurately handle complex open-ended responses.

Recommendations for Implementation

To maximize the system's effectiveness, institutions should adopt several key strategies. First, implementing mixed formats, such as the 70% multiple-choice and 30% short-answer structure, can provide an optimal balance between automation and the depth of analysis required for meaningful assessment. Second, reliance on short-answer-only formats should be minimized, as these formats increase cognitive strain and reduce user satisfaction, as seen in cases like H4 and H8. Finally, engagement strategies should be tailored to each user cluster. For low-perception users, simplification and clear communication of benefits should be prioritized. For moderate users, the focus should be on incentivizing usage and continuously improving the experience. For high-perception users, empowering them as advocates and deepening their system expertise will help sustain engagement and promote broader acceptance.

By addressing these recommendations, institutions can significantly enhance perceptions of the system's usefulness, ease of use, and overall engagement. This will ultimately lead to higher adoption rates and more effective integration of AI-assisted grading systems in educational environments.

Conclusion:

The study highlights the transformative potential of artificial intelligence (AI) and large language models (LLMs) in higher education, particularly in the domain of grading and assessment. While AI-assisted grading systems offer unparalleled efficiency, consistency, and scalability, student perceptions underscore the critical need for maintaining human oversight to ensure reliability, fairness, and contextual accuracy. This balance between automation and human involvement can foster trust and enhance the credibility of AI systems among students and educators alike.

A key insight from the study is the efficacy of mixed assessment formats—those that combine multiple-choice and short-answer questions. This balanced approach optimally leverages AI's strengths in processing structured responses while addressing its limitations in interpreting complex, open-ended answers. The findings indicate that such formats not only improve students' perceived usefulness (PU) and perceived ease of use (PEOU) but also significantly enhance attitudes toward use (ATU) and behavioral intention (BI). These outcomes highlight the importance of thoughtful exam design in maximizing the effectiveness of AI-assisted grading systems.

The application of the Technology Acceptance Model (TAM) reveals that perceived usefulness and ease of use remain pivotal in shaping students' acceptance of AI-based tools. However, future research should expand TAM to incorporate additional constructs, such as trust, fairness, and transparency, which are increasingly important in the context of AI adoption. Exploring these dimensions will provide a more holistic understanding of user acceptance and address potential concerns about bias, equity, and ethical considerations in automated assessments. Furthermore, longitudinal studies should examine the long-term impacts of AI on learning outcomes, student engagement, and academic performance to validate the broader benefits of these technologies in education.

To foster sustainable adoption of AI-assisted grading systems, institutions should prioritize the following measures:

1. **Balanced Assessment Designs:** Implement mixed formats that integrate 70% multiple-choice and 30% short-answer questions. This structure optimizes the advantages of automation while preserving the depth and critical thinking assessment associated with short-answer formats. Institutions should also minimize reliance on purely short-answer assessments to reduce cognitive strain and avoid undermining user satisfaction.
2. **Comprehensive Training Programs:** Provide targeted training for both students and educators to enhance their understanding of AI capabilities and limitations. Training should include demonstrations of how AI grading systems work, the rationale behind their use, and strategies for effectively interpreting AI-generated feedback.
3. **Human Oversight and Transparency:** Maintain a strong human presence in the grading process to validate AI outputs, particularly for subjective or complex responses. Clearly communicate how AI systems make decisions to build trust and dispel misconceptions about their accuracy and fairness.

4. **Ethical and Equity Considerations:** Address ethical concerns by implementing safeguards against potential biases in AI grading. Institutions should ensure that AI systems are tested across diverse student populations and provide mechanisms for students to challenge or appeal grading decisions.

5. **Feedback Mechanisms:** Establish robust feedback loops to gather student and faculty insights on AI-assisted grading. Regular updates and system improvements driven by this feedback will help refine AI tools and demonstrate responsiveness to user needs.

6. **Recognition and Incentives:** Encourage early adopters of AI-assisted grading to serve as ambassadors by recognizing their contributions and offering professional development opportunities. These ambassadors can advocate for broader adoption and help others navigate the transition to AI-supported systems.

By implementing these recommendations, institutions can create a collaborative ecosystem where AI acts as a tool to enhance learning experiences, streamline assessment processes, and support educational excellence. Future innovations in AI-assisted grading should be guided by a commitment to fairness, transparency, and inclusivity, ensuring that these technologies are embraced as integral components of a modern, equitable educational framework.

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APPENDIX 1

Example of Scenarios

Hypothetical Scenario 1: TA Grading - Exam with All Multiple-Choice Questions

You are a student enrolled in A Principles of Marketing class at your university. The semester is coming to an end, and you have an upcoming final exam. Here's what you need to know about the exam:

Exam Format:

- Type of Questions: The exam consists of 50 multiple-choice questions.
- Content Coverage: The questions cover all the topics taught throughout the semester, including marketing principles, consumer behavior, market segmentation, and marketing strategies.
- Time Limit: You have 2 hours to complete the exam.

Grading Method:

- Graded by: The exam will be manually graded by the Teaching Assistant (TA) assigned to your class.
- Grading Process:
 - The TA will review each of your selected answers.
 - They will ensure that any ambiguities or unclear markings are interpreted correctly.
 - If there are any issues with a question (e.g., ambiguities or errors), the TA can make adjustments during grading.

Instructions Provided:

- Answer Sheet: You will mark your answers on a provided answer sheet.
- Marking Answers:
 - Clearly circle or fill in the letter corresponding to your chosen option (A, B, C, or D).
 - You may write notes or calculations in the margins if needed.
- Flexibility: The TA understands that mistakes can happen, so if you need to change an answer, you can erase or cross it out and mark your new choice clearly.
- Additional Notes:
 - If you have any concerns or comments about a question, you can write them on the exam paper.
 - The TA will consider your notes during grading.

Hypothetical Scenario 2: ChatGPT Grading - Exam with All Multiple-Choice Questions

You are a student enrolled in A Principles of Marketing class at your university. The semester is coming to an end, and you have an upcoming final exam. Here's what you need to know about the exam:

Exam Format:

- Type of Questions: The exam consists of 50 multiple-choice questions.
- Content Coverage: The questions cover all the topics taught throughout the semester, including marketing principles, consumer behavior, market segmentation, and marketing strategies.
- Time Limit: You have 2 hours to complete the exam.

Grading Method:

- Graded by: The exam will be automatically graded by ChatGPT, an AI language model.
- Grading Process:
 - The AI system will scan your answers and automatically score your exam.
 - Grading is based strictly on the answers provided without human intervention.
 - The system ensures quick turnaround times for results.

Instructions Provided:

- Answer Sheet: You will mark your answers on a provided answer sheet designed for AI grading.
- Marking Answers:
 - Only write the letter (A, B, C, or D) corresponding to your chosen option in the designated space for each question.
 - Do not include any additional markings, notes, or symbols near your answers.
- Formatting Requirements:
 - Use clear, legible handwriting.
 - Ensure that your letters are capitalized and not cursive.
 - Do not cross out or erase; if you need to change an answer, clearly indicate your final choice.
- Additional Notes:
 - Any deviations from the instructions may result in the AI misreading your answers.
 - The system cannot interpret additional notes or comments.